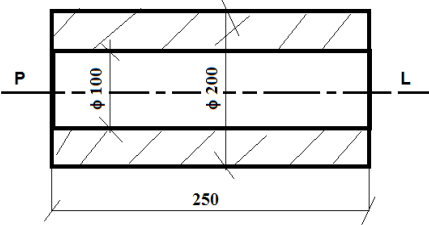


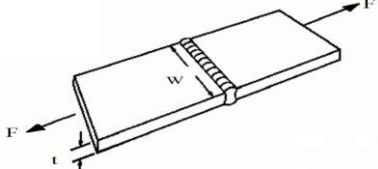
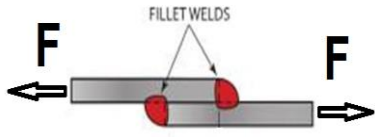
SEC.		I.D		أسم الطالب / (باللغة العربية ويخط واضح)
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1	The cast is smaller than the pattern by:	2	The feeding functions are:	3	The casting pattern is made of:
A	Shrinkage and Machining All.	A	Indicator of mold filling, escape of hot air from the mold, compensate shrinkage during solidification.	A	Foam
B	Machining Allowance	B	Deliver the molten metal for mold filling, escape of molten from the mold, compensate shrinkage during solidification.	B	Any material
C	Shrinkage Allowance	C	Indicator of mold filling, escape of hot air from the mold, compensate shrinkage during melting.	C	Metal
D	Draft Allowance	D	compensate shrinkage during solidification only.	D	Wood
4	Casting pattern is defined as:	5	The Casting Mold is defined as:	6	Permanent pattern is....
A	Positive replica of the cast	A	Positive replica of the cast	A	Not removed after mold formation
B	Positive replica of the Final product	B	Positive replica of the Final product	B	Removed after mold formation
C	Negative replica of the cast	C	Negative replica of the cast	C	Removed after final product formation
D	Negative replica of the Final product	D	Negative replica of Final product	D	Removed after cast formation
7	Casting mold in is made of:	8	The gating system function is:	9	Use Metal casting is beneficial in
A	Wood	A	Delivers the molten metal to mold.	A	Solid and thin hollow products.
B	Foam	B	Delivers the molten metal to cast.	B	Hollow products only.
C	Sand	C	Delivers molten metal to pattern.	C	Solid products only.
D	Metal or Sand	D	Delivers molten to final product.	D	Thin hollow products only
10	Yield in metal casting refers to:	11	Continuous casting products are:	12	The disposable pattern is made of:
A	$W(\text{Cast}) / W(\text{molten})$.	A	Shaped castings only.	A	Metal
B	$W(\text{Pattern}) / W(\text{molten})$.	B	Wrought alloys castings only.	B	Wood
C	$W(\text{Final Product}) / W(\text{molten})$.	C	Wrought and Shaped castings.	C	Foam
D	$W(\text{molten}) / W(\text{Cast})$.	D	Ingot castings only.	D	Sand
13	The Physically bonded sand is	14	The Final product is smaller than the pattern by the allowance	15	Continuous casting mold is made of
A	Sand + Binder.	A	Shrinkage all.	A	Copper.
B	Sand + Binder + Activator.	B	Shrinkage and Machining all.	B	Sand.
C	Sand + water	C	Machining all.	C	Tool Steel
D	Sand + Thermosetting Resin.	D	Shrinkage and Machining and draft.	D	Wood
16	Continuous casting yield is =	17	Select the suitable Flask Dim.	18	Select the suitable Cast Dim.
A	Almost 100 %.	A	500 X 400 X 200 mm	A	$D_{\text{outer}} = 204, D_{\text{inner}} = 96, H = 254$ mm
B	As high as 90 %.	B	400 X 300 X 150 mm	B	$D_{\text{outer}} = 202, D_{\text{inner}} = 96, H = 252$ mm
C	$W(\text{molten}) / W(\text{Cast})$.	C	450 X 400 X 150 mm	C	$D_{\text{outer}} = 196, D_{\text{inner}} = 96, H = 254$ mm
D	60 %.	D	450 X 300 X 150 mm	D	$D_{\text{outer}} = 204, D_{\text{inner}} = 104, H = 254$ mm
Q 17 & 18, 19, 20 For the shown final product, knowing that it will be machined in all inner and outer surfaces, taken into consideration: Machining allowance = 2 mm, shrinkage allowance = 2 mm/m.				19	Select the suitable Core Dim.
Draft:				A	$D = 96.2$ mm, $H = 254.5$ mm
				B	$D = 102.2$ mm, $H = 354.5$ mm
				C	$D = 96.2$ mm, $H = 354.5$ mm
				D	$D = 96.2$ mm, $H = 304.5$ mm
		20	Select the suitable Pattern Dim.		
		A	$D_{\text{outer}} = 202.4, D_{\text{core print}} = 96.2, H = 254.5$ mm		
		B	$D_{\text{outer}} = 204.4, D_{\text{core print}} = 96.2, H = 254.5$ mm		
		C	$D_{\text{outer}} = 204.4, D_{\text{core print}} = 95.8, H = 254.5$ mm		
		D	$D_{\text{outer}} = 204.4, D_{\text{core print}} = 96.2, H = 254$ mm		

With best wishes for success

Dr. Mohamed Atef Sayed
Dr. Tarek M. Moussa

SEC.		B.N.		أسم الطالب / (باللغة العربية وبخط واضح)
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21	Oxygen cylinder fitting is:	22	DCEP indicates:	23	DCEN indicates:
A	Right-handed threads.	A	AC current.	A	AC current.
B	Left-handed threads.	B	DC current reversed polarity.	B	DC current reversed polarity.
C	Any of the above.	C	DC current straight polarity.	C	DC current straight polarity.
D	None of the above.	D	None of the above	D	None of the above
24	Code [E6010] indicates:	25	E6010 consumable is used for:	26	The form of welding flux is:
A	600 MPa. Strength.	A	GTAW	A	Granular or powder.
B	60 Psi. Strength.	B	GMAW	B	Electrode coating.
C	60 Kg/mm ² Strength.	C	FCAW	C	Electrode core.
D	None of the above.	D	None of the above.	D	Any of the above.
27	In SAW process:	28	In GMAW process	29	ESW is used to weld:
A	Coated electrodes are used.	A	Un-fused flux is collected by vacuum.	A	Thin plates in vertical position.
B	Shielding Gas is used.	B	Slag removal is needed	B	Thick Plates in flat position.
C	Carbon electrode is used.	C	Tungsten electrode is used.	C	Thin plates in flat position.
D	None of the above.	D	None of the above.	D	None of the above.
30	High voltage & current lead to:	31	E70-T-1 consumable is used for:	32	ER70-S-6 consumable is used for:
A	Spray transfer mode.	A	SMAW	A	SMAW
B	Globular transfer mode.	B	GMAW	B	SAW
C	Short circuiting.	C	PAW	C	FCAW
D	None of the above	D	None of the above.	D	None of the above.
33	In FCAW process	34	Double Orifice Nozzle is used in	35	Key-Hole Welding Technique can be achieved in
A	Carbon electrode is used	A	SMAW	A	GTAW
B	Tungsten electrode is used.	B	GMAW	B	GMAW
C	Double Orifice Nozzle is used	C	FCAW	C	FCAW
D	None of the above.	D	PAW	D	None of the above.
36	Permanent electrode is used for:	37	Continuous wire electrode is used for:	38	In CAW process, the used electrode is:
A	ESW & EGW.	A	PAW.	A	Consumable steel electrode.
B	SMAW.	B	CAW.	B	Permanent tungsten electrode.
C	FCAW & GMAW.	C	GTAW.	C	Continuous wire electrode.
D	None of the above.	D	None of the above.	D	None of the above.
Calculate [F] the force needed to break the Joint, Two plates, Steel (St.52), [t = 5 mm & W = 20 mm] SMAW Electrode E7018 Note that: [1 Ksi = 6.89 MPa] ($\tau = 0.6 \sigma$)					39 The force needed to break the Joint= A 12500 [N] B 24100 [N] C 48230 [N] D None of the above.
Calculate [F] the force needed to break the Joint, Two plates, Steel (St.52), [t = 5 mm & W = 20 mm] SMAW Electrode E7018 Note that: [1 Ksi = 6.89 MPa] ($\tau = 0.6 \sigma$)					40 The force needed to break the Joint= A 43138.2 [N] B 12941.4 [N] C 25882.9 [N] D None of the above.

With best wishes for success

Dr. Ahmed Farid A.G. Youssef